MAM2000W Module 2LA: Linear Algebra

Chapter 1 **Some answers to the exercises** 2023

Section 1.1

1. 20 tons of each type of fuel should be manufactured.

2. The equations are:

$$\begin{aligned} j_{k+1} &= 0.33a_k \\ s_{k+1} &= 0.18j_k \\ a_{k+1} &= 0.71j_k + 0.94a_k. \end{aligned}$$

Section 1.2

Note: Questions 1,and 4 must be answered for matrices of arbitrary sizes; it is not in order to assume that they are all 2×2.

1. In each case, show that the entry in the i-th row and j-th column on the left hand side is equal to the entry in the i-th row and j-th column on the right hand side.

2. (a) $A = \begin{pmatrix} -5 & 2 & 3 \\ -2 & 0 & 2 \\ -4 & 2 & 2 \end{pmatrix}$

   (b) $A = \begin{pmatrix} -2-\lambda & 2 & 3 \\ -2 & 3-\lambda & 2 \\ -4 & 2 & 5-\lambda \end{pmatrix}$

3. Check your answers with OCTAVE or Wolfram Alpha.
   (e) BA is not defined.

4. Let A be an $m \times n$ matrix, and B be a $n \times p$ matrix, and $C = AB$.
   (a) Let row ℓ in matrix A have only zeros. For each integer j such that $1 \leq j \leq p$, we have

$$c_{\ell j} = \sum_{k=1}^{n} a_{\ell k} b_{kj} = 0,$$

   because $a_{\ell k} = 0$ for all k. So row ℓ in the matrix AB also has zeroes only.
   (b) Use a similar argument to (a) above.

5. (a) True.
   If $kA = O$ and $k \neq 0$, then $A = \frac{1}{k}(kA) = \frac{1}{k}O = O$.

[There are other ways of writing this out; this is certainly one of the fastest. If the logic isn't clear to you, we are using the fact that, for any propositions P, Q, R, the statement $P \Rightarrow Q \vee R$ is logically equivalent to $P \wedge \neg Q \Rightarrow R$.]
(b) False.
As a counter-example you could take $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

6. As a counter-example, let $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$. Then
$(A+B)^2 = \begin{pmatrix} 4 & -3 \\ 0 & 1 \end{pmatrix}$, but $A^2 + 2AB + B^2 = \begin{pmatrix} 3 & -2 \\ -1 & 2 \end{pmatrix}$. A correct expansion is:

$$\begin{aligned} (A+B)^2 &= (A+B)(A+B) \\ &= A(A+B) + B(A+B) \\ &= A^2 + AB + BA + B^2. \end{aligned}$$

Here we use the left and right distributive laws. Because matrix multiplication is not commutative, we cannot assume that $AB = BA$.

7. Let $A = (a_{ij})$, $B = (b_{ij})$ and $C = (c_{ij})$. Then

$$\begin{aligned} \text{The } (i,j)\text{-th entry of } A(B+C) &= \sum_{k=1}^{n} [a_{ik}(b_{kj} + c_{kj})] \\ &= \sum_{k=1}^{n} [(a_{ik}b_{kj}) + (a_{ik}c_{kj})] \\ &= \sum_{k=1}^{n} a_{ik}b_{kj} + \sum_{k=1}^{n} a_{ik}c_{kj} \\ &= \text{the } (i,j)\text{-th entry of } AB + AC. \end{aligned}$$

Notice that we are using the fact that the addition and multiplication of real numbers satisfies the distributive law (where?) and the commutative law (where?).

8. Use a similar strategy to the one used in Question 7.

9. Use $(A+B)^T = A^T + B^T$.

10. True. We have $\mathbf{0} = A^2 = AA^T$. Now use the fact that each diagonal element of AA^T must be 0.

11. Check, using the definitions of symmetric and skew-symmetric.

12. (b) $B+C=2A$.

13. We want to show that, if A is an $n\times n$ matrix, then A can be written uniquely in the form $A=S+K$, where S is symmetric and K is skew-symmetric.
Let $S=\frac{1}{2}(A+A^T)$ and $K=\frac{1}{2}(A-A^T)$. It is clear that $S+K=\frac{1}{2}(A+A^T)+\frac{1}{2}(A-A^T)=\frac{1}{2}A+\frac{1}{2}A^T+\frac{1}{2}A-\frac{1}{2}A^T=A$. Question 11 shows that that S is symmetric and K is skew-symmetric. So we know that we can express A in the required way; what remains is to show that this is the *only* possible way. For this, suppose that $A=S_1+K_1=S_2+K_2$, where S_1,S_2 are both symmetric, and K_1,K_2 are both skew-symmetric. Then $S_1-S_2=K_2-K_1$ and S_1-S_2 is symmetric and K_2-K_1 is skew-symmetric. Is it possible for a matrix to be symmetric *and* skew-symmetric? Let's see: If B is both, then $B^T=B=-B$, so $B=O$, a zero matrix. So $S_1-S_2=O$ and $K_2-K_1=O$, so $S_1=S_2$ and $K_1=K_2$, as required.

Section 1.3

1. (a) $\left(\dfrac{7}{8},-\dfrac{3}{2},0\right)$
   (b) $\left(-\dfrac{1}{6},\dfrac{3}{2},1,2\right)$
   (c) Inconsistent.

Section 1.4

1. (a) Yes (b) No (c) Yes (d) No (e) No (f) Yes

2. (a) $(x,y,z)=(-5,0,-2)+t(\dfrac{1}{2},1,0)$ for $t\in\mathbb{R}$.
   (b) $(x_1,x_2,x_3,x_4)=(\dfrac{18}{4},0,-2,0)+t(\dfrac{5}{4},0,-1,1)+s(-\dfrac{3}{4},1,0,0)$, where $s,t\in\mathbb{R}$.
   (c) $(x,y,z,w)=\left(-\dfrac{10}{3},-\dfrac{5}{2},1,0\right)+t\left(\dfrac{2}{3},-\dfrac{3}{4},-\dfrac{1}{2},1\right)$, where $t\in\mathbb{R}$.
   (d) $(x_1,x_2,x_3,x_4)=\left(\dfrac{1}{2}t-8,t,-2,3\right)$, where $t\in\mathbb{R}$.
   (e) $(x,y,z,u,v)=(t,\dfrac{s}{4}-2,s,3,0)$, where $t,s\in\mathbb{R}$.

3. (c) $(u,v,w,x,y)=\left(\dfrac{3}{2},0,2,5,0\right)+t\left(\dfrac{3}{2},1,0,0,0\right)+s\left(-\dfrac{1}{2},0,-3,-3,1\right)$, where $s,t\in\mathbb{R}$.
   (d) $(x_1,x_2,x_3,x_4)=\left(\dfrac{53}{45}+\dfrac{32}{9}s,\dfrac{9}{5}+8s,\dfrac{1}{15}(4+20s),s\right)$, where $s\in\mathbb{R}$.

4. Use the Linear Algebra Toolkit to check your answer.

5. (a) $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

   (b) $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

6. (a) F (b) F (c) T (d) T

Section 1.5

1. (a) Let I be the $n \times n$ identity matrix. In order to have $AB = I$, we would need A to be $n \times p$ and B to be $p \times n$, for some natural number p. On the other hand, to have $BA = I$, we would need A to be $q \times n$ and B to be $n \times q$, for some natural number q. So to have $AB = I = BA$, we need $n = p = q$, which makes A and B both $n \times n$.
   (b) False. For example, let A be any $n \times n$ matrix such that $A \neq I_n$, and $B = I_n$. (For instance you could take $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.) Then A and B are both square, and $AB = BA$, but $A \neq B^{-1}$.

2. Show that if $ad - bc = 0$, then the matrix is singular.

3. First we show that, for any invertible matrix B, $(B^T)^{-1} = (B^{-1})^T$: This means that we must show that the inverse of B^T is $(B^{-1})^T$; so we multiply together $B^T.(B^{-1})^T = (B^{-1}B)^T = I^T = I$ (see Theorem 1.4.18). Similarly, $(B^{-1})^T.B^T = I$. This gives $(B^T)^{-1} = (B^{-1})^T$.
   Now suppose that A is symmetric and invertible (i.e. is non-singular); so $A = A^T$. Then $(A^{-1})^T = (A^T)^{-1} = A^{-1}$, (make sure you understand each of these steps) so A^{-1} is indeed symmetric.

Section 1.6

1. (a) $e^{-1} = e$ in this case.
   (b) Add 2 times the first row to the third row.
   (c) Multiply the third row by $\frac{3}{2}$.
   (d) Add -7 times the fourth row to the second row.

2. None of the given matrices is elementary.

3. You should get $e(B) = EB$, $e(C) = EC$.

4. (a) $E_1 = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$, $E_2 = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$
   (b) $E_2E_1 = A^{-1}$
   (c) $A = E_1^{-1}E_2^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$

5. Use OCTAVE to check your answers. (Once you have entered A, the command "inv(A)" will give the inverse of A if it exists.) Alternativelybuse Wolfram Alpha to check your answers.

6. Use $AB = I$ to show that B is non-singular, hence invertible. Then $AB = I$ implies $B^{-1} = A$.

7. See Question 5 (b),(c) of Tutorial 3.

8. Hint: If $\mathbf{x} = \mathbf{y}$ is any solution of $A\mathbf{x} = \mathbf{b}$, show that $\mathbf{y} - \mathbf{c}$ is a solution of $A\mathbf{x} = \mathbf{0}$.

9. (a) Let $\mathbf{b} = (b_1, b_2, b_3)^T$.
   (i) $b_2 + b_3 = 0$ (ii) $b_2 + b_3 \neq 0$ (iii) Not possible.
   (b) Singular.

10. (a) True. Hint for proof: See Question 3 of Tutorial 4.
    (b) False. See tutorial 4, Question 4.

11. There are several possible answers for this question, depending on the elementary row operations you use to reduce A to the identity matrix. See Question 4 of Tutorial 3 for a solution of a similar question.

12. B is not necessarily invertible. It will only be invertible if all the diagonal elements in U are non-zero.

Section 1.7

1. (a) 0 (b) 52 (c) 0 (d) -123

2. (a) -5 (b) -1 (c) 1 (d) 0

3. -120

4. -216

5. (a) -189 (b) $-\dfrac{1}{7}$ (c) $-\dfrac{8}{7}$ (d) $-\dfrac{1}{56}$ (e) 7

6. The eigenvalues are 1 and 0.

7. In Tutorial 3, Question 5(c) we proved that if A and B are square of the same size and AB is invertible, then A and B are invertible. Consider the contrapositive.